

USE CASE “Place order”

I. Use Case “Place order”

1. Use case code

UC001

2. Brief description

This use case describes the sequential interaction between the customer and the AIMS software when the customer decides to finalize their shopping cart and place an order for media products.

3. Actors

Customer

4. Preconditions

- There must be at least one available physical media product (Book, CD, DVD, etc.) currently placed in the customer's active shopping cart.

5. Basic Flow of Events

1. Customer proceeds to place an order.
2. Software checks whether the available stock is sufficient to fulfill the customer's demand.
3. Software asks customers to set up delivery information.
4. Customer enters the delivery information (**see Table 2**).
5. Software checks the input information.
6. Software calculates and displays the delivery fee (**see Table 4**).
7. Customer proceeds to pay the order.
8. Software calls the “Pay order”
9. Software displays invoice screen (**see Table 4**).
10. Software displays a QR code for the customer to scan as the default payment method (see Table 3).
11. Customer completes the payment.
12. Software displays transaction information (see Table 5).
13. Software records the payment transaction information and the successfully paid order.
14. Software puts the order in a pending processing state.
15. Software sends invoice and payment transaction information to the customer's email.

6. Alternative flows

Table 1 – Alternative flows of events for UC place order

No	Location	Condition	Action	Resume location
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1	At step 2	If the stock is insufficient	System prompts the customer that the store is out of the product	Use case end
2	At step 5	If there are any required fields left blank or invalid information	Software asks customers to update the information.	Resumes at Step 4
3	At Step 6	If customers update the delivery address	Software will recalculate and display the delivery fee.	Resumes at Step 7
4	At Step 8	If customers adjust the delivery method or the items they wish to purchase	Software recalculates the delivery fees and updates the corresponding invoice.	Resumes at Step 8
5	At Step 10	If the customer does not wish to pay by QR code	Customer may switch to an alternative payment method using a credit card.	Resumes at Step 10
6	At any step	If customers go back to any step or exit the software during the ordering process	Software returns to the previous step or exits the process entirely.	Corresponding step or Use case ends

7. Input data

Table 2 – Input data of delivery information

No	Data fields	Description	Mandatory	Valid condition	Example
1	Receiver Name	Full name of the person receiving the package	Yes	Not blank	Thanh Huu Dat
2	Phone	Contact number	Yes	- 10 digits - May contains separator “.” and/or space and/or “-” in the middle of the phone number (one separator type), interleave with digits	0383841363 0914362222

				- Starts with "0"	
3	Province	Delivery region or city	Yes	Chose from a drop-down list	Ha Noi
4	Address	Detailed street and house information	Yes	Not blank	3 Bo De street
5	Shipping Instructions	Additional notes for the shipper	NO		The house which is opposite the church

Table 3 – Input data of payment

No	Data fields	Description	Mandatory	Valid condition	Example
1	Payment method	Choice between default QR Code (VietQR) or alternative Credit Card (PayPal)	Yes	Valid payment option	Credit Card
2	Card details	Required if the customer switches to Credit card	Yes	Valid card number, expiry, CVV	1234-5678-9012-3456

8. Output data

Table 4 – Output data of invoice screen

No	Data field	Description	Display format	Example
1	Title	Title of the media product	Text	"DVD phim mưa đỏ."
2	Price	Price of product	-Comma for thousands separator -Positive integer - Right alignment	12,500
3	Quantity	Number of units of product	-Positive number -Right	2

			alignment	
4	Amount	Total price of the corresponding product	-Comma for thousands separators -Positive integer -Right alignment	25,000
5	Subtotal	Total amount of all products	-Comma for thousands separators -Positive integer -Right alignment	150,000
6	Shipping Fee	Calculated delivery cost based on weight/location	-Comma for thousands separators -Positive integer -Right alignment	30,000
7	Total	Final price	Comma for thousands separators -Positive integer -Right alignment	180,000

Table 5– Output data of transaction

No	Data field	Description	Display format	Example
1	Customer name		Text	Thanh Huu Dat
2	Phone number		Positive integer	0914362222
3	Province		Text	Ha Noi
4	Address		Text	3 Bo De street
5	Total amount		Right alignment Vietnamese currency (VNĐ)	180,000

			Vietnamese locale	
6	Transaction ID		Positive integer	12345
7	Transaction content		Text	You have a success transaction at 05/10/2023
8	Transaction date		dd/mm/yyyy	05/10/2023

9. Postconditions

- The successfully paid order is recorded and is in a pending processing state.
- The payment transaction information is recorded.
- The customer receives an email with the invoice and payment transaction information.
- The quantity in stock of the purchased products is automatically updated.